Long Term Portfolios in Asset Pricing Models

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Introduction

Purpose of the paper

- Use empirical facts on long term assets to exam asset pricing models

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Motivation

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- Use empirical facts on long term assets to exam asset pricing models

Motivation

- Long term assets carry substantial information on long run discount rate and cash flow news
  - → Long bond returns help characterize long run discount factor (Alvarez and Jermann, 2005)
  - → Spread of Long equity and Long bond measures long run cash flow news (Ai and Fu, 2026)

Two primary assets

Long term assets

- Portfolios with cash flows realized in the distant future

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My analysis focuses on

1. Long term bond
   - Pays 1 at $t+n$ with large n

$$B_{t,n} = E_t\left(\frac{S_{t+n}}{S_t}\right)$$

2. Long term equity
   - Pays $D_{t,n}$ at $t+n$ with large n

$$P_{t,n} = E_t\left(\frac{S_{t+n}}{S_t} D_{t+n}\right)$$

Outline

1. Document important stylized facts on long term assets
   - Equity term structure
   - Risk characteristics of long equity
   - Predictability
   - Long term risk premium
   - Volatility of infinite-maturity excess returns

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   - Long run risks
   - External habit

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1. Document important stylized facts on long term assets
   - Equity term structure
   - Risk characteristics of long equity
   - Predictability
   - Long term risk premium
   - Volatility of infinite-maturity excess returns
2. Evaluate the properties of leading asset pricing models
   - Long run risks
   - External habit
3. (In the future) Develop a model better capturing long term asset returns

Related Literature

Long Term Pricing

Alvarez and Jermann (2005), Hansen and Scheinkman (2009), Ross (2015), Borovicka, Hansen, Scheinkman (2016)

Term Structure

Van Binsbergen, Brandt and Koijen (2012), Van Binsbergen and Koijen (2017), Bansal, Miller, Song and Yaron (2021)

Primitives in LRR (BY04)

Key channel: Low frequency movements and time-varying uncertainty in consumption growth

Cash flow (Y = C or D)

$$\begin{aligned}
\Delta c_{t+1} &= \mu + x_t + \sigma_t \eta_{t+1} \\
x_{t+1} &= \rho x_t + \varphi_e \sigma_t e_{t+1} \\
\sigma_{t+1}^2 &= \sigma^2 + \nu_1 \left(\sigma_t^2 - \sigma^2\right) + \sigma_\omega \omega_{t+1} \\
\Delta d_{t+1} &= \mu_d + \phi x_t + \varphi_d \sigma_t u_{t+1}
\end{aligned}$$

Stochastic discount factor (S)

$$\Delta s_{t+1} = \theta \log \delta - \frac{\theta}{\psi} \Delta c_{t+1} + (\theta - 1) r_{w,t+1}$$

Primitives in HAB (CC99)

Key channel: Time varying risk aversion and hence risk premia

Cash flow (Y = C or D)

$$\Delta c_{t+1} = \mu + \sigma_\nu \nu_{t+1}$$

$$\Delta d_{t+1} = \mu + \sigma_\omega \omega_{t+1}, \quad \text{Corr}\left(\omega_t, \nu_t\right) = \rho$$

Stochastic discount factor (S)

$$\Delta s_{t+1} = \text{const} + \gamma(1-\phi)m_t - \gamma(1+\lambda(m_t))\sigma_\nu \nu_{t+1}$$

$$m_{t+1} = (1-\phi)\bar{m} + \phi m_t + \lambda(m_t)\sigma_\nu \nu_{t+1}$$

Operator Approach of Hansen and Sheinkman

- A vector of Markov state variables X with LOM:

$$X_{t+1} = \phi_\pi \left(X_t, \Delta W_{t+1} \right)$$

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$$\ln H_{t+1} - \ln H_t = \kappa_H \left(X_t, \Delta W_{t+1} \right)$$

  - E.g. SDF (S_t), cash flow (Y_t) and their multiple ($S_t Y_t$)

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  - E.g. SDF (S_t), cash flow (Y_t) and their multiple ($S_t Y_t$)

- Decomposition:

$$\frac{H_{t+1}}{H_t} = \underbrace{e^{\rho_H}}_{\text{Constant Drift}} \underbrace{\frac{\mathbf{e}_H^{-1}\left(X_{t+1}\right)}{\mathbf{e}_H^{-1}\left(X_t\right)}}_{\text{Stationary}} \underbrace{\frac{\widehat{H}_{t+1}}{\widehat{H}_t}}_{\text{Martingale}}$$

Impulse response functions

Different asset pricing models

- Match similar set of moments of aggregate market and risk free rate

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- But with different dynamic structures for Y and S

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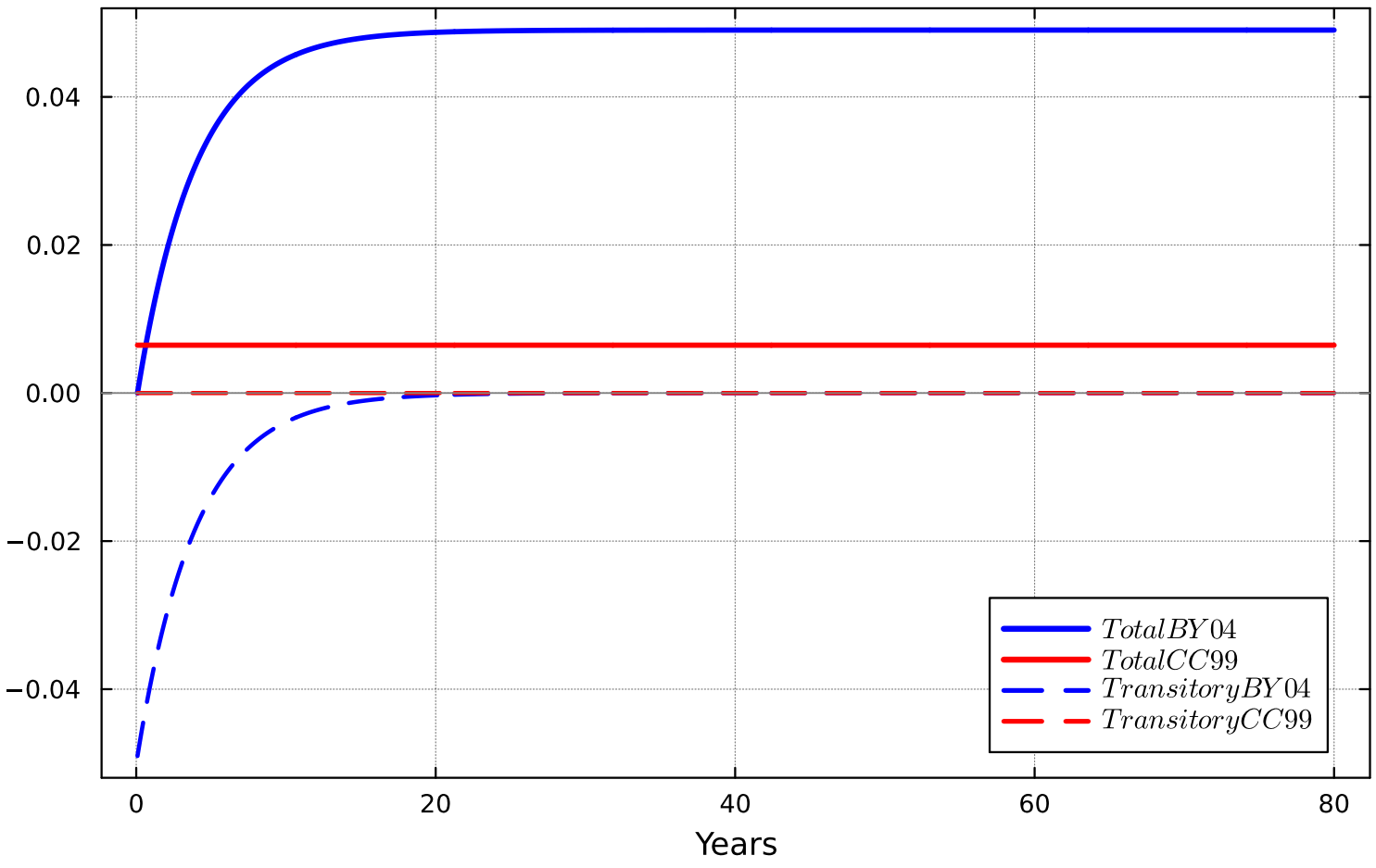
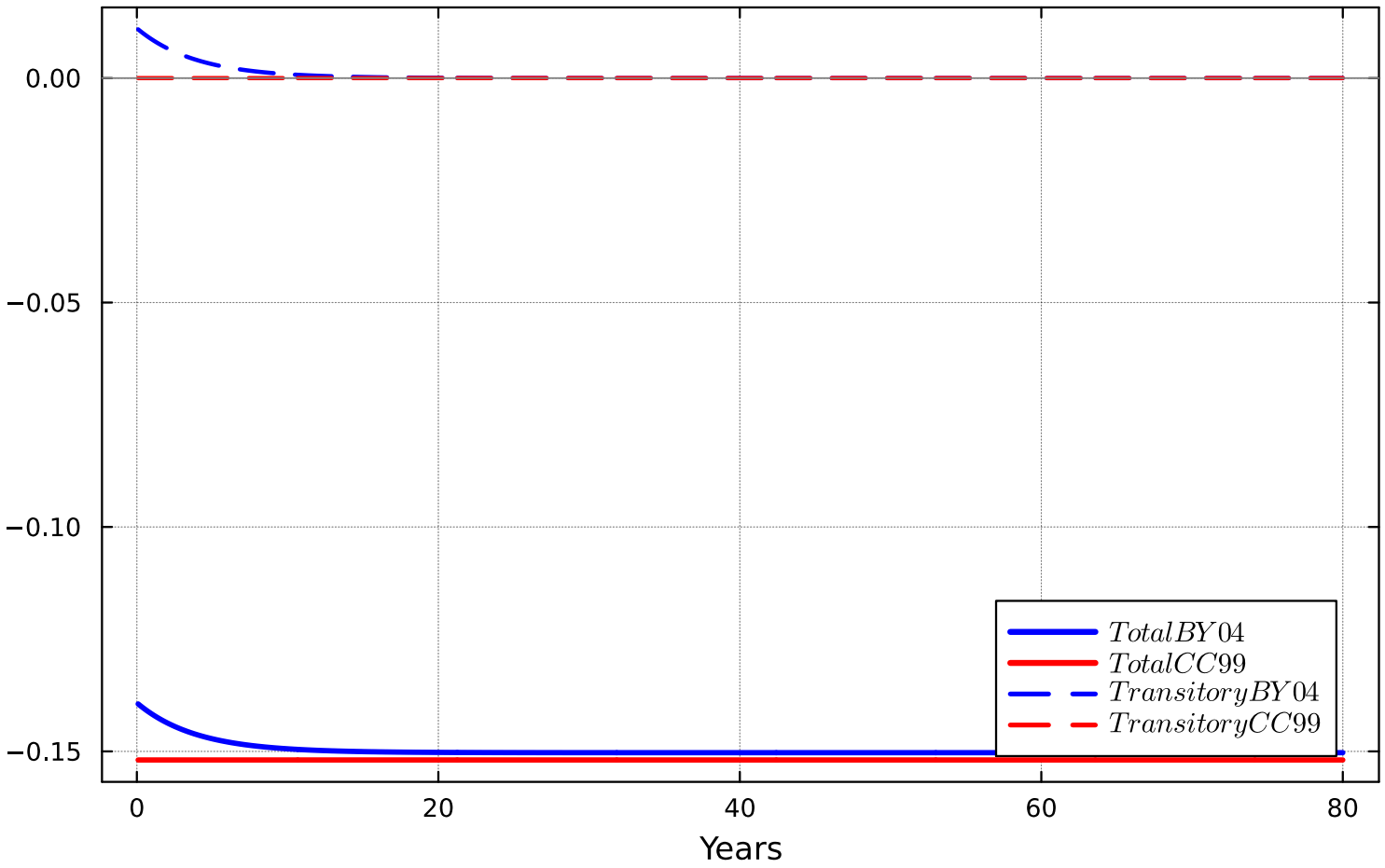
To see that

- Compare the IRFs of $H \in \{D, S, SD\}$ in BY04 and CC99
  1. $IRF_H(h) = \log \mathbb{E}\left(\frac{H_{t+h}}{H_t} \middle| X_{t+1}^1\right) - \log \mathbb{E}\left(\frac{H_{t+h}}{H_t} \middle| X_{t+1}^0\right)$
  2. $IRF_{\mathbf{e}^{-1}}(h) = \log \mathbb{E}\left(\frac{\mathbf{e}_H(X_t)}{\mathbf{e}_H(X_{t+h})} \middle| X_{t+1}^1\right) - \log \mathbb{E}\left(\frac{\mathbf{e}_H(X_t)}{\mathbf{e}_H(X_{t+h})} \middle| X_{t+1}^0\right)$
  3. $IRF_{\hat{H}}(h) = \log \mathbb{E}\left(\frac{\hat{H}_{t+h}}{\hat{H}_t} \middle| X_{t+1}^1\right) - \log \mathbb{E}\left(\frac{\hat{H}_{t+h}}{\hat{H}_t} \middle| X_{t+1}^0\right)$
- One standard deviation x shock to BY04 and ν shock to CC99

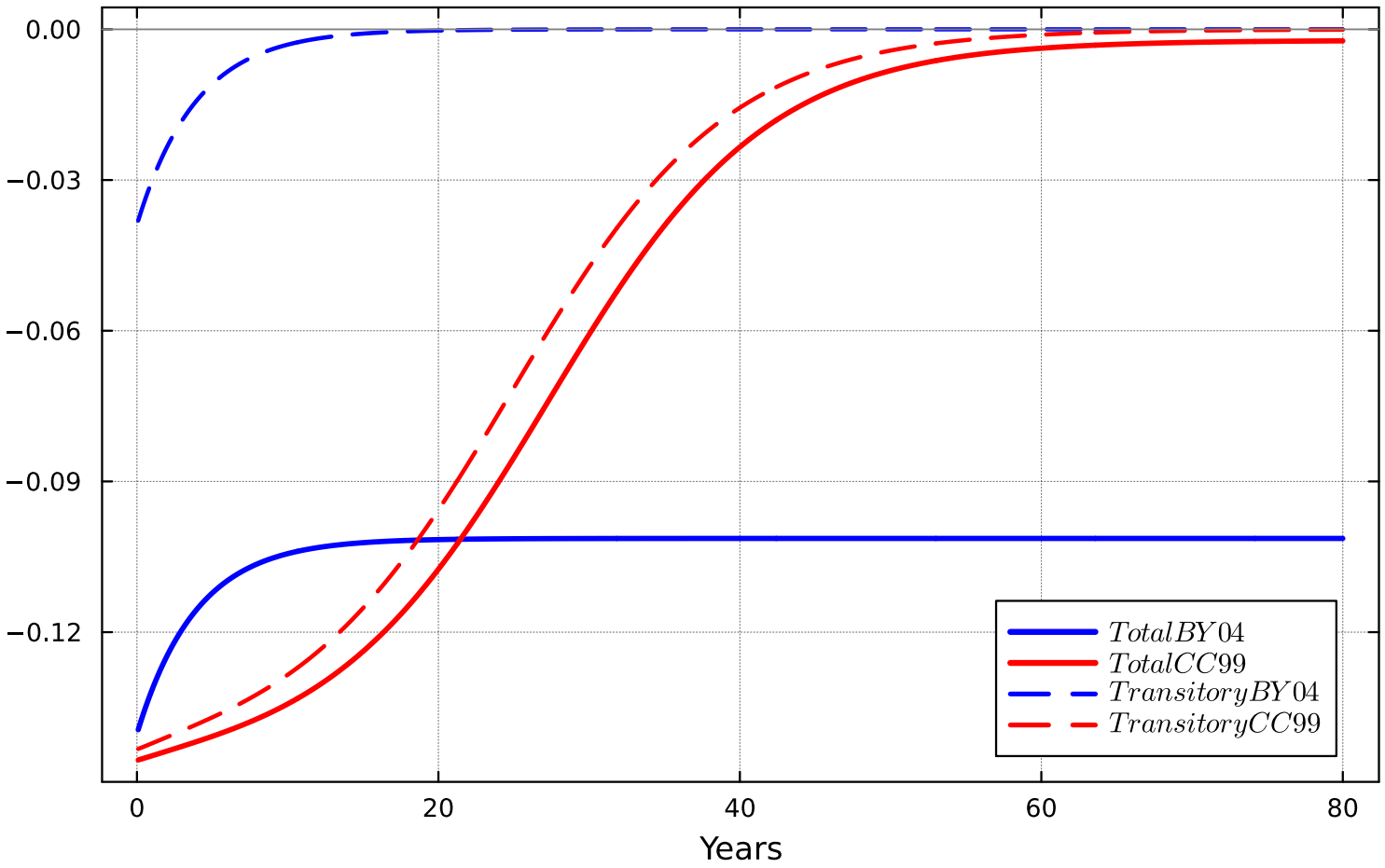
IRFs of H = D


0.04
0.02
0.00
−0.02
−0.04
0
20
40
60
80
Years
$TotalBY04$
$TotalCC99$
$TransitoryBY04$
$TransitoryCC99$

IRFs of H = S


0.00
−0.05
−0.10
−0.15
0
20
40
60
80
Years
$TotalBY04$
$TotalCC99$
$TransitoryBY04$
$TransitoryCC99$

IRFs of H = SD


0.00
−0.03
−0.06
−0.09
−0.12
0
20
40
60
80
Years
$TotalBY04$
$TotalCC99$
$TransitoryBY04$
$TransitoryCC99$

Long term asset returns

- Returns of long term bond and equity:

$$R^{\infty}_{B,t+1} = e^{r_S} \frac{\mathbf{e}_S(X_{t+1})}{\mathbf{e}_S(X_t)} \qquad R^{\infty}_{E,t+1} = e^{r_{SY}} \frac{\mathbf{e}_{SY}(X_{t+1})Y_{t+1}}{\mathbf{e}_{SY}(X_t)Y_t}$$

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- What can long term assets returns reveal?

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- What can long term assets returns reveal?
  1. Transitory
     - Properties of $\mathbf{e}_S(X)$ and $\mathbf{e}_{SY}(X)Y$ (new)

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- What can long term assets returns reveal?
  1. Transitory
     - Properties of $\mathbf{e}_S(X)$ and $\mathbf{e}_{SY}(X)Y$ (new)
  2. Permanet
     - $L\left(\frac{\widehat{S}_{t+1}}{\widehat{S}_t}\right) - L\left(\frac{\widehat{SY}_{t+1}}{\widehat{SY}_t}\right) = \mathbb{E}\left(r^{\infty}_{Y,t+1} - r^{\infty}_{B,t+1}\right)$ (new)
     - $L\left(\frac{\widehat{S}_{t+1}}{\widehat{S}_t}\right) \geq \mathbb{E}\left(r_{t+1} - r^{\infty}_{B,t+1}\right)$ (AJ05)
     - $(\mathbb{E}_{t+1} - \mathbb{E}_t) \log \frac{\hat{Y}_{t+1}}{\hat{Y}_t} \approx r^{\infty}_{Y,t+1} - r^{\infty}_{B,t+1}$ (AF26)

Proxies for long term assets

Long bond

- Treasury real bond with longest maturity (20Y TIPS)

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Long equity

- Portfolios of non-dividend paying stocks
  1. No dividend payment up to date t
  2. Stated no dividend in the foreseeable future on 10K annual reports

Proxies for long term assets

Long bond

- Treasury real bond with longest maturity (20Y TIPS)

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Sample periods

- Fact 1 to Fact 3 on long equity and market: 1965-2024
- Fact 4 to Fact 5 on long equity and long real bond: 1999-2024

Equity term structure

Fact 1

Data: LE return is lower and more volatile than the market, hence a lower Sharpe ratio.
BY04:

CC99:

Mean and volatility			
	$E(r)\%$	$\sigma(r)\%$	Sharpe
Long Equity	4.73	25.39	0.30
Market	6.16	15.69	0.44
BY04 LE			
BY04 MKT			
CC99 LE			
CC99 MKT			

Equity term structure

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BY04: Slightly upward sloping equity term structure. Long equity has too high mean and too low volatility.
CC99:

Mean and volatility			
	$E(r)\%$	$\sigma(r)\%$	Sharpe
Long Equity	4.73	25.39	0.30
Market	6.16	15.69	0.44
BY04 LE	7.65	18.26	0.37
BY04 MKT	6.70	16.46	0.33
CC99 LE			
CC99 MKT			

Equity term structure

Fact 1

Data: LE return is lower and more volatile than the market, hence a lower Sharpe ratio.
BY04: Slightly upward sloping equity term structure. Long equity has too high mean and too low volatility.
CC99: Largely upward sloping equity term structure. Long equity return way too high mean and volatility.

Mean and volatility			
	$E(r)\%$	$\sigma(r)\%$	Sharpe
Long Equity	4.73	25.39	0.30
Market	6.16	15.69	0.44
BY04 LE	7.65	18.26	0.37
BY04 MKT	6.70	16.46	0.33
CC99 LE	14.09	52.70	0.49
CC99 MKT	7.30	20.21	0.41

Riskiness of Long Equity

Fact 2

Data: CAPM works with $\beta = 1.43$ and an insignificant $\alpha = -2.5\%$
BY04:
CC99:

CAPM				
	$\alpha_{CAPM}(\%)$	β_{MKT}		
Long Equity	-2.5	1.43^{***}		
BY04 LE				
CC99 LE				
FF 3-factors				
	$\alpha_{FF}(\%)$	β_{MKT}	β_{SMB}	β_{HML}
Long Equity	-0.72	1.23^{***}	0.68^{***}	-0.51^{***}

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CAPM				
	$\alpha_{CAPM}(\%)$	β_{MKT}		
Long Equity	-2.5	1.43^{***}		
BY04 LE	0.7^{***}	1.10^{***}		
CC99 LE				
FF 3-factors				
	$\alpha_{FF}(\%)$	β_{MKT}	β_{SMB}	β_{HML}
Long Equity	-0.72	1.23^{***}	0.68^{***}	-0.51^{***}

Riskiness of Long Equity

Fact 2

Data: CAPM works with $\beta = 1.43$ and an insignificant $\alpha = -2.5\%$
BY04: CAPM fails. β is close but α is not.
CC99: CAPM fails. Both β and α are too large.

CAPM				
	$\alpha_{CAPM}(\%)$	β_{MKT}		
Long Equity	-2.5	1.43^{***}		
BY04 LE	0.7^{***}	1.10^{***}		
CC99 LE	6.4^{***}	2.51^{***}		
FF 3-factors				
	$\alpha_{FF}(\%)$	β_{MKT}	β_{SMB}	β_{HML}
Long Equity	-0.72	1.23^{***}	0.68^{***}	-0.51^{***}

Predictability

Fact 3

Data: No predictability found for both long equity and market for 1965-2024
BY04:

CC99:

	Long Equity ($r_E - r_f$)			MKT($r_M - r_f$)		
	1Y	3Y	5Y	1Y	3Y	5Y
Data						
β_{pd}	-0.12	-0.13	-0.09	-0.06	-0.10	-0.15
R^2	0.02	0.01	0.00	0.02	0.02	0.03
BY04						
β_{pd}						
R^2						
CC99						
β_{pd}						
R^2						

Predictability

Fact 3

Data: No predictability found for both long equity and market for 1965-2024
BY04: Consistent with the data pattern of this period. Predictability is similar for long equity and market
CC99:

	Long Equity ($r_E - r_f$)			MKT($r_M - r_f$)		
	1Y	3Y	5Y	1Y	3Y	5Y
Data						
β_{pd}	−0.12	−0.13	−0.09	−0.06	−0.10	−0.15
R^2	0.02	0.01	0.00	0.02	0.02	0.03
BY04						
β_{pd}	−0.05	−0.16	−0.25	−0.04	−0.12	−0.20
R^2	0.01	0.02	0.03	0.01	0.02	0.03
CC99						
β_{pd}						
R^2						

Predictability

Fact 3

Data: No predictability found for both long equity and market for 1965-2024
BY04: Consistent with the data pattern of this period. Predictability is similar for long equity and market
CC99: Inconsistent. Also, β_{pd} is larger for long equity than for the market.

	Long Equity $(r_E - r_f)$			MKT$(r_M - r_f)$		
	1Y	3Y	5Y	1Y	3Y	5Y
Data						
β_{pd}	−0.12	−0.13	−0.09	−0.06	−0.10	−0.15
R^2	0.02	0.01	0.00	0.02	0.02	0.03
BY04						
β_{pd}	−0.05	−0.16	−0.25	−0.04	−0.12	−0.20
R^2	0.01	0.02	0.03	0.01	0.02	0.03
CC99						
β_{pd}	−0.58	−1.49	−2.10	−0.24	−0.60	−0.86
R^2	0.08	0.20	0.29	0.09	0.21	0.28

Long term risk premium

Fact 4

Data: The average return of long bond is 2.92% and long term risk premium is 1.76%
BY04:
CC99:

	$\mathbb{E}\left(r^{\infty}_{B,t+1}\right)$	$\mathbb{E}\left(r^{\infty}_{Y,t+1} - r^{\infty}_{B,t+1}\right)$
Data	2.92%	1.76%
BY04 LE		
CC99 LE		

- $L\left(\frac{\widehat{S}_{t+1}}{\widehat{S}_t}\right) - L\left(\frac{\widehat{SY}_{t+1}}{\widehat{SY}_t}\right) = \mathbb{E}\left(r^{\infty}_{Y,t+1} - r^{\infty}_{B,t+1}\right)$

Long term risk premium

Fact 4

Data: The average return of long bond is 2.92% and long term risk premium is 1.76%
BY04: Low long bond return and too high long term risk premium
CC99:

	$\mathbb{E}\left(r^{\infty}_{B,t+1}\right)$	$\mathbb{E}\left(r^{\infty}_{Y,t+1} - r^{\infty}_{B,t+1}\right)$
Data	2.92%	1.76%
BY04 LE	0.41%	7.23%
CC99 LE		

- $L\left(\frac{\widehat{S}_{t+1}}{\widehat{S}_t}\right) - L\left(\frac{\widehat{SY}_{t+1}}{\widehat{SY}_t}\right) = \mathbb{E}\left(r^{\infty}_{Y,t+1} - r^{\infty}_{B,t+1}\right)$

Long term risk premium

Fact 4

Data: The average return of long bond is 2.92% and long term risk premium is 1.76%
BY04: Low long bond return and too high long term risk premium
CC99: Low long bond return and way too high long term risk premium

	$\mathbb{E}\left(r^{\infty}_{B,t+1}\right)$	$\mathbb{E}\left(r^{\infty}_{Y,t+1} - r^{\infty}_{B,t+1}\right)$
Data	2.92%	1.76%
BY04 LE	0.41%	7.23%
CC99 LE	0.94%	13.15%

- $L\left(\frac{\widehat{S}_{t+1}}{\widehat{S}_t}\right) - L\left(\frac{\widehat{SY}_{t+1}}{\widehat{SY}_t}\right) = \mathbb{E}\left(r^{\infty}_{Y,t+1} - r^{\infty}_{B,t+1}\right)$

Vol of Long Term Excess Returns

Fact 5

Data: Vol of the long term excess returns is 25.4%
BY04:
CC99:

	Data	BY04 LE	CC99 LE
$\sigma(r_Y^\infty - r_B^\infty)$	25.4%		
$\sigma\left(\ln \frac{\hat{Y}_{t+1}}{\hat{Y}_t}\right)$			

Vol of Long Term Excess Returns

Fact 5

Data: Vol of the long term excess returns is 25.4%
BY04: This vol matches data. Permanent component dominates.
CC99:

	Data	BY04 LE	CC99 LE
$\sigma(r_Y^\infty - r_B^\infty)$	25.4%	21.64%	
$\sigma\left(\ln \frac{\hat{Y}_{t+1}}{\hat{Y}_t}\right)$		21.02%	

Vol of Long Term Excess Returns

Fact 5

Data: Vol of the long term excess returns is 25.4%
BY04: This vol matches data. Permanent component dominates.
CC99: This vol is too large. Permanent component plays less important role.

	Data	BY04 LE	CC99 LE
$\sigma(r_Y^\infty - r_B^\infty)$	25.4%	21.64%	52.7%
$\sigma\left(\ln \frac{\hat{Y}_{t+1}}{\hat{Y}_t}\right)$		21.02%	11.2%